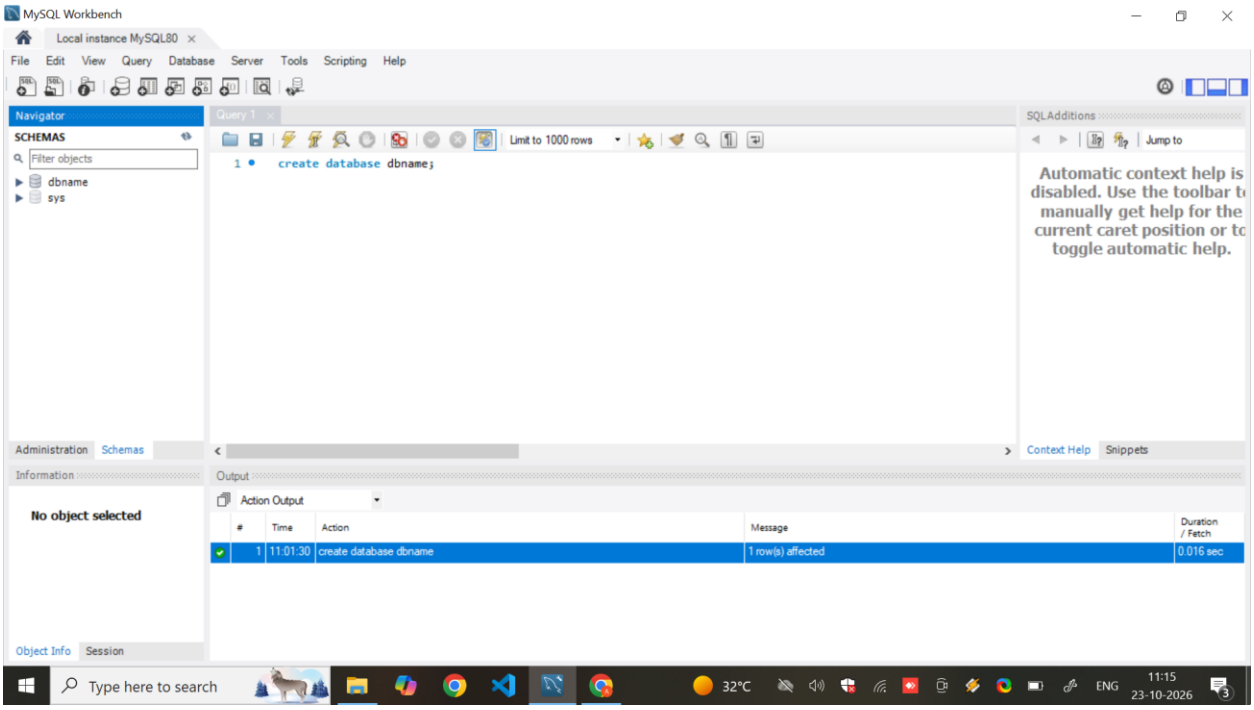
Session 1

Q. 1) Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.



Q. 2) Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

Ans. Create table users

(user_id int(11) primary key auto increament,

user_name varchar(255),

email varchar(100),

followers_count int(11));

Q. 3) Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

Ans. Insert into users (user_name, email, followers_count)

values

('riya_shah', 'riya@example.com', 1500),

('arjun_mehta', 'arjun@example.com', 2000),

('priya_patel', 'priya@example.com', 5000);

Q. 4) Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

Ans. Create table posts

(post_id int(11) primary key, user_id int(11), caption varchar(255), post_date date,
foreign key(post_id) references users(user_id) on update cascade on delete cascade);

Q. 5) Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

Ans.

1. MySQL

MySQL is a popular database used for websites and web applications.
It is commonly used with PHP, WordPress, and e-commerce websites.
It is easy to learn and suitable for small to large applications.

2. PostgreSQL

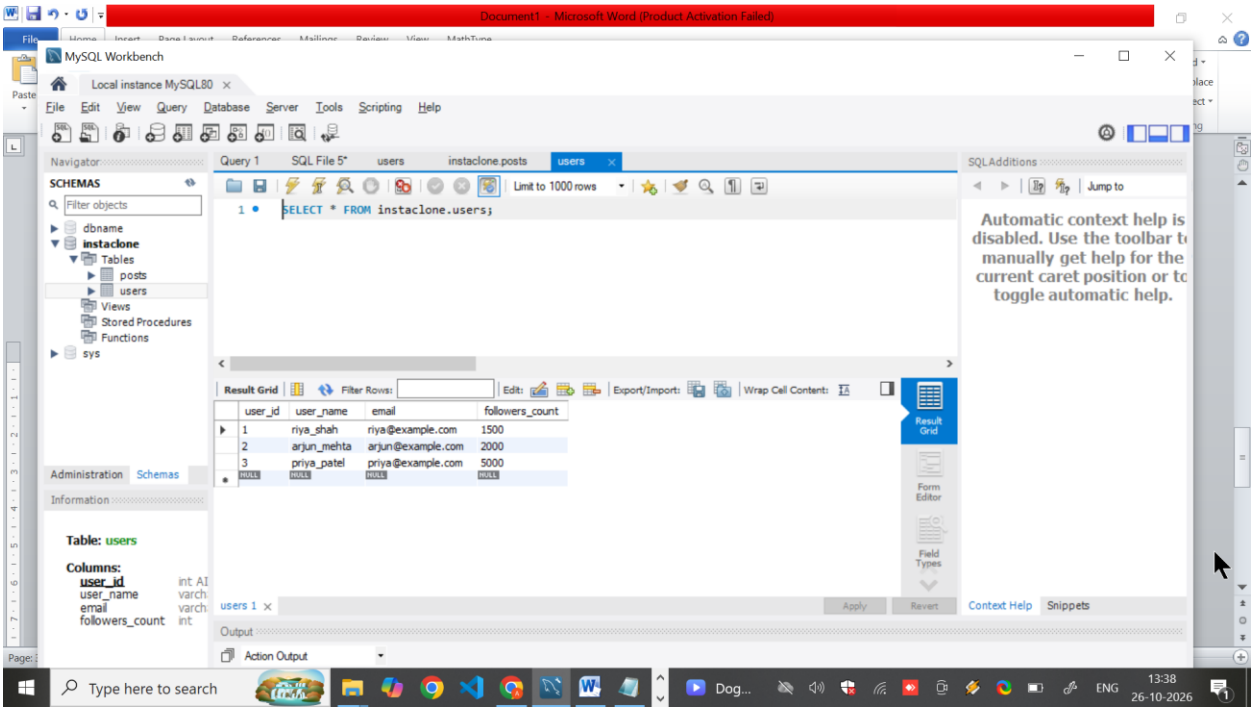
PostgreSQL is an advanced open-source database known for reliability and accuracy.
It is used in web applications, data analytics, and applications that handle complex data.
It is suitable for projects requiring advanced queries and strong data integrity.

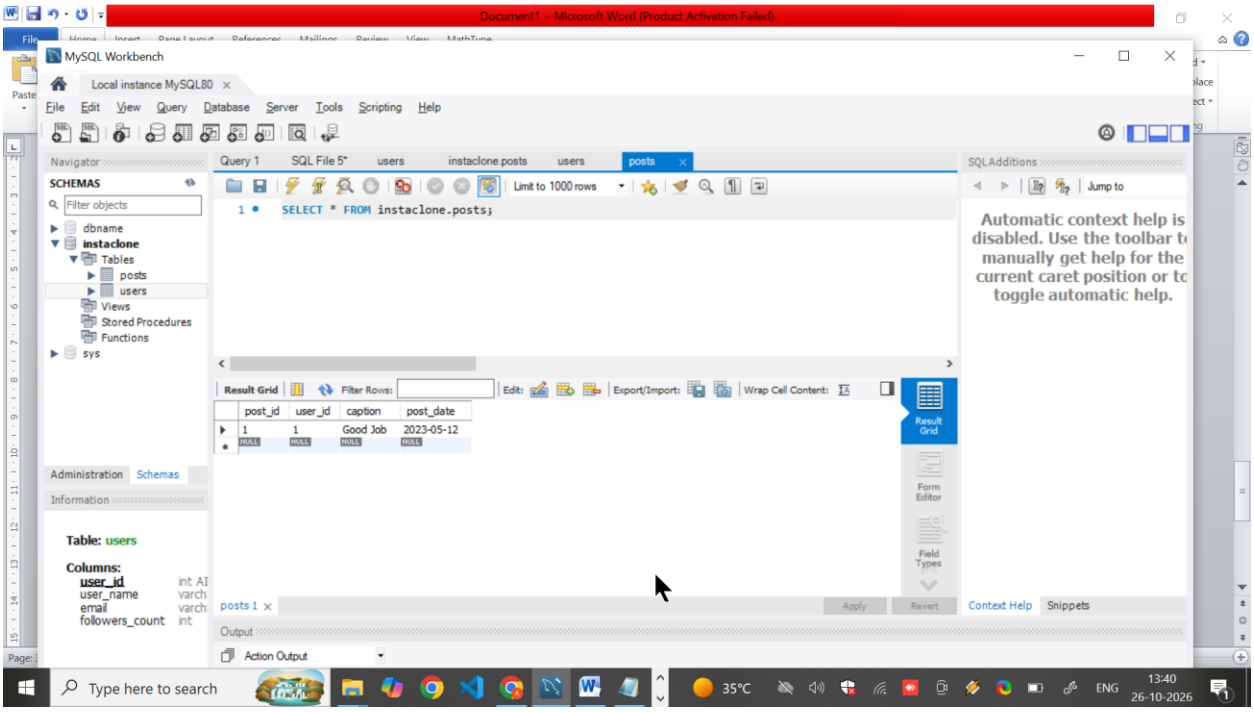
3. Oracle Database

Oracle is a powerful database mainly used in large enterprise systems.
It is commonly used by banks, hospitals, and large companies to manage huge amounts of data.
It offers advanced security, reliability, and transaction management.

4. SQLite

SQLite is a lightweight database that stores data in a single file without requiring a separate server.
It is commonly used in mobile apps, desktop applications, and embedded systems.
It is ideal for small applications and offline data storage.


Document1 - Microsoft Word (Product Activation Failed)
MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Query 1 SQL File 5* users instaclone.posts users
SELECT * FROM instaclone.users;
SCHEMAS
dbname
instaclone
Tables
posts
users
Views
Stored Procedures
Functions
sys
Result Grid
user_id user_name email followers_count
1 riya_shah riya@example.com 1500
2 arjun_mehta arjun@example.com 2000
3 priya_patel priya@example.com 5000
Table: users
Columns:
user_id int AI
user_name varch
email varch
followers_count int
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or tc toggle automatic help.
Type here to search
ENG
13:38
26-10-2026


Document1 - Microsoft Word (Product Activation Failed)
MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Query 1 SQL File 5* users instaclone.posts users posts
SELECT * FROM instaclone.posts;
SCHEMAS
dbname
instaclone
Tables
posts
users
Views
Stored Procedures
Functions
sys
Result Grid
post_id user_id caption post_date
1 1 Good Job 2023-05-12
Table: users
Columns:
user_id int AI
user_name varch
email varch
followers_count int
Automatic context help is disabled. Use the toolbar t manually get help for the current caret position or tc toggle automatic help.
Type here to search
35°C
ENG
13:40
26-10-2026